

TOPAZ: Noise-Resilient Quantum Circuit Architectures via Geometric Wells in the Parameter Landscape

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Abstract

Realizing the practical utility of NISQ-era quantum computing requires executing complex operations within the strict limitations of physical qubit coherence times. This challenge is most pronounced in variational quantum algorithms (VQAs), where static, unoptimized ansatz structures introduce severe operational vulnerabilities: they run redundant gates that actively accumulate device errors, distorting the loss landscape and accelerating the onset of barren plateaus. Traditional post-processing error mitigation strategies modify gate schedules or stretch pulses, but they fail to alter these underlying, noise-susceptible circuit topologies. We introduce TOPAZ (Topology OPTimization Ansatz with Zoning), a universal hardware-resilient circuit synthesis framework that moves beyond traditional ansatz limitations by porting classical structural topology optimization directly to quantum computing. Paralleling the Solid Isotropic Material with Penalization (SIMP) method in structural mechanics, TOPAZ models an unoptimized circuit execution path as a continuous structural mesh. The framework maps the target routine to a local parametric Trotter expression (LPTE), parameterizing the network using individual gate intensities (ρ) and timings (τ). Parameter zoning decomposes these heterogeneous design variables into decoupled tracks: a dual Method of Moving Asymptotes (dual-MMA) solver updates intensities and timings under separate asymptotes, while a discrete cut prunes underutilized entangling edges to drastically compress physical circuit depth. Evaluated via a tensor-dot Kraus engine ($O(2^{2N})$ per term, yielding a 3838 $\times$ simulation speedup), TOPAZ provides passive run-time error resilience across arbitrary quantum algorithms. Benchmarked on the 8-qubit simulator FakeFz, TOPAZ yields compiled meshes that are 34% shallower at negligible fidelity cost ($\Delta\text{DM} = -0.0065$). Physical execution on the `ibm_fez` (Heron r2) processor demonstrates a one-way state fidelity of $57.58\% \pm 1.11\%$ and a two-way Loschmidt echo of $17.27\% \pm 0.60\%$ —an 11.14 $\times$ improvement over product-Trotter baselines that completely outperforms active zero-noise extrapolation. These results demonstrate that TOPAZ provides a scalable synthesis pipeline by driving quantum operations into sparse, structural wells that natively absorb physical device noise.

Keywords: TOPAZ, topology optimisation, Parametric Trotter Expression, zoning, noise-resilient circuits, Method of Moving Asymptotes, structural well.

Code and data. Qiskit version X, Aer Y, MMA implementation Z. Hardware job ids TBD. Optimised (ρ, τ) parameters, per-seed outputs, and plotting scripts will be deposited at a public archive upon acceptance.

Variational quantum algorithms [6, 12] represent the dominant paradigm for near-term quantum utility, yet they remain trapped by a fundamental trade-off: an ansatz must be deep enough to express the target state, but shallow enough to execute within physical coherence times (T_2). Conventional variational circuits rely on static, unoptimized topologies—such as Hardware-Efficient Ansätze (HEA) [21] or Hamiltonian Variational Ansätze (HVA) [22]—which execute redundant two-qubit gates that actively accumulate device error, distort the loss landscape, and accelerate the onset of barren plateaus [29]. Standard quantum error mitigation (QEM) strategies—including Zero-Noise Extrapolation (ZNE) [7], Clifford Data Regression (CDR) [8], and Virtual Distillation (VD) [9]—address this failure only after execution [16]. They stretch pulses, fit decay curves, or amplify state copies, but leave the underlying error-accumulating circuit topology unaltered: a near-zero rotation angle that is never pruned still compiles to noisy physical CNOTs.

In this work, we resolve this bottleneck prior to execution. We introduce **TOPAZ (Topology OPTimization Ansatz with**

Zoning), a classical design framework that ports structural topology optimization—specifically the Solid Isotropic Material with Penalization (SIMP) method [4, 5]—to quantum circuit synthesis. By modeling an unoptimized circuit execution path as a continuous structural mesh, TOPAZ maps target operations to a Local Parametric Trotter Expression (LPTE) [3], parameterizing the network with independent gate intensities (ρ) and timings (τ). Decoupling these heterogeneous variables via parameter zoning, a dual Method of Moving Asymptotes (dual-MMA) solver optimizes continuous parameters while a discrete threshold cut prunes underutilized entangling edges, collapsing circuit depth.

Related approaches modify the operator set in orthogonal directions. ADAPT-VQE [23] *grows* generators from the energy gradient; TOPAZ *shrinks* (ρ, τ) under a noise objective and never adds an edge. Molecular VQE on superconducting devices [24, 25] and the NISQ survey [26] treat mitigation as post-processing; we stack ZNE, CDR and VD on a circuit that has already been designed (Sec. 5.7), and we do not claim a chemistry record. The geometric language for the

basins TOPAZ lands in—structural wells—is developed in a companion note [1]; this manuscript uses only what the design loop needs.

The primary contributions of this work are:

1. **Continuous Operator Relaxation.** We formalize a non-unitary continuous relaxation for quantum circuits via polar retraction ($P_U(A) = A(A^\dagger A)^{-1/2}$), proving that the Fréchet derivative strips non-unitary stretching perturbations (Lemma 1) and that unital noise leaves the critical points of the loss landscape invariant (Lemma 2).
2. **TOPAZ Synthesis Pipeline.** We establish an end-to-end circuit optimization protocol combining LPTE with dual-MMA, demonstrating a 34% reduction in two-qubit depth with negligible fidelity loss ($\Delta\text{DM} = -0.0065$) via automated topological pruning.
3. **Accelerated Contraction Infrastructure.** We construct a tensor-dot Kraus contraction engine operating at $O(2^{2N})$ per term, achieving a 3,838× wall-clock speedup over standard superoperator methods at $N = 10$.
4. **Hardware Demonstration on 156-Qubit Heron.** Executed on the physical `ibm_fez` (Heron r2) processor, TOPAZ achieves a two-way Loschmidt echo of $17.27\% \pm 0.60\%$ —an **11.14× improvement** over standard product Trotter ($1.55\% \pm 0.28\%$) and 3× above active gate-stretched ZNE (5.78%).

1 Continuous Topological Interpretation

CTI [1] supplies the vocabulary and the geometric picture that TOPAZ operationalises. We summarise only what is needed for the rest of the paper.

1.1 Ambient space and polar projection

Standard quantum circuits operate in the unitary group $U(d) = \{U \in \text{Mat}_d(\mathbb{C}) : U^\dagger U = I\}$, a compact Lie group of dimension d^2 where $d = 2^N$. CTI embeds $U(d)$ as a constrained submanifold inside the full space of linear operators $\text{Mat}_d(\mathbb{C}) \cong \mathbb{C}^{d^2}$. A continuous circuit operator $A(\lambda)$ maps $\mathbb{R}^M \rightarrow \text{Mat}_d(\mathbb{C})$ without the unitarity constraint, allowing gradients to flow around barriers that would block a strictly unitary optimizer.

To execute operators on hardware, we project back to $U(d)$ via the polar decomposition. Every invertible $A \in \text{GL}(d, \mathbb{C})$ decomposes uniquely as $A = QP$ with Q unitary and $P = (A^\dagger A)^{1/2}$ positive definite [20, Sec. 8.1]:

$$P_U(A) = A(A^\dagger A)^{-1/2}. \quad (1)$$

Q is the unique closest unitary to A in the Frobenius norm. This projection — **unitary regularization** — pins singular values back to 1.0, smoothing sharp wrinkles in the loss landscape and stabilizing gradient signals. The decomposition is unique if and only if A is invertible; in LPTE the per-pair operator $B_{(i,i+1)}$ is lower-bounded by the clipping floor $\max(\rho, 10^{-6})$, so $\sigma_{\min}(B)$ remains strictly positive and the SVD polar factor $U = B(B^\dagger B)^{-1/2}$ is uniquely defined, guaranteeing that the projection is deterministic and differentiable on the active block.

1.2 Entanglement as Manifold Enlargement via Dynamical Lie Algebras

The accessible state manifold of an ansatz — the space of physical states it can possibly reach — is rigorously dictated by the Dynamical Lie Algebra (DLA) generated by its operators, and its expressibility is quantified by DLA growth [31]. Without entanglement, operations are restricted to localized rotations, and the generators are local Pauli operators (e.g., X_1, Z_1, X_2, Z_2). Because these operators act on disjoint qubits, they commute across qubit boundaries (e.g., $[X_1, X_2] = 0$). Their Lie closure remains small, yielding a DLA of dimension only $3N$. This severely restricts the optimizer’s ability to navigate the Hilbert space.

Introducing entangling parameters generates non-local terms (e.g., $X_1 X_2$). Because local and non-local generators do not commute, taking their commutators generates entirely new, non-local Pauli strings; for example, $[X_1 X_2, Y_2 Y_3] = 2i X_1 Z_2 Y_3$. This new operator can then be commuted further, causing the DLA to grow and expanding the reachable state manifold.

By contrast, a generic Hardware-Efficient Ansatz (HEA) mixes local and arbitrary non-local generators freely, causing its DLA to rapidly close to the full unitary algebra $\mathfrak{su}(2^N)$ (dimension $4^N - 1$); this unstructured growth is associated with barren plateaus [29], where gradient variance vanishes exponentially. The LPTE ansatz restricts its generators to per-pair $\{XX, YY, ZZ\}$ triples. A single LPTE block spans an Abelian subalgebra ($[XX, YY] = 0$), so its local parameter space is small; chained blocks grow the reachable manifold via cross-pair commutators without closing to $\mathfrak{su}(2^N)$ at our scales. We do not claim a barren-plateau theorem from this algebraic picture — our Hessian measurements are consistency checks, not proofs. This structured, block-by-block growth is what makes local pruning safe: removing an entire chain removes only its own block algebra, leaving the reachable manifold of the remaining blocks unchanged.

1.3 The Structural Well and Topological Invariance

Environmental noise (T_1, T_2 , readout error) acts as a continuous vector field pulling the state landscape away from the target unitary manifold. In standard circuits this deforms the loss landscape, shifting minima and creating barren plateaus. Instead of correcting the output, TOPAZ searches for a **structural well** — a region of the parameter manifold where the noisy output is intrinsically robust. We formalise this intuition with one definition, two lemmas that hold exactly, and two propositions whose assumptions are stated explicitly; the cross-noise-model evidence itself is empirical.

Definition 1 (Parameter space). The optimization domain is $\mathcal{X} = [0, 1]^M \times [0, \tau_{\max}]^M$, where M is the number of entangling pairs. This space is closed and bounded, hence compact. The loss function $L : \mathcal{X} \rightarrow \mathbb{R}$ is continuous, defined as $L(\rho, \tau) = 1 - |\langle \psi_{\text{ideal}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2$.

Remark 1 (Well family: assumed versus observed). By the extreme value theorem, the continuous loss L attains a global minimum on the compact domain $\mathcal{X}$; existence of some minimiser is all that this argument buys. That the landscape hosts *multiple* wells is an experimental observation (different seeds converge to different floors), not a derivation. For the rest of

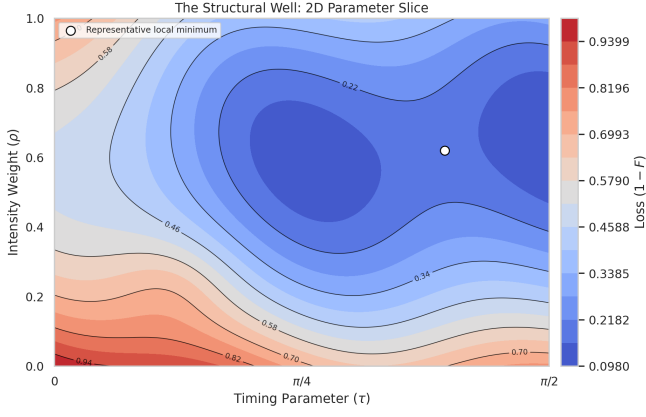


Figure 1. The structural well on FakeFaz, projected onto a 2D (ρ, τ) slice. Colour encodes the loss $L = 1 - F$; solid contours are sublevel sets $S_c(L)$ as defined in Remark 1.

this section, a well is a connected component $B_c(x_i^*)$ of a sublevel set $S_c(L) = \{(\rho, \tau) \mid L(\rho, \tau) \leq c\}$ containing a local minimum x_i^* with $\nabla L(x_i^*) = 0$ and $\nabla^2 L(x_i^*) \succ 0$, and $\mathcal{W}$ denotes the collection of these basins. That different seeds land in different wells follows from pruning: each seed freezes a different sparse subset of parameters.

Lemma 1 (Classical Manifold Stability via Fréchet Derivative). *To show noise cannot push the optimizer out of these wells, we must first examine the local geometry that governs the classical optimization path. Let the non-unitary PTE operator be $B = QP$ (polar decomposition), where Q is unitary and P is Hermitian positive-definite. A small perturbation $\delta B = \delta Q P + Q \delta P$ decomposes into a rotation (δQ , tangent to the unitary manifold) and a stretch (δP , normal to it). Using the unitary constraint $Q^\dagger Q = I$, we define $\Omega = Q^\dagger \delta Q$, which satisfies $\Omega + \Omega^\dagger = 0$ (anti-Hermitian). Solving the Sylvester equation for the near-unitary case yields the Fréchet derivative of the polar projection:*

$$\delta Q = \frac{1}{2}(\delta B - B \delta B^\dagger B).$$

If the optimizer takes a step that is purely non-unitary stretching, $\delta B = B \delta P = \delta P B$. Substituting this into the equation above yields:

$$\delta Q = \frac{1}{2}(B \delta P - B \delta P) = 0.$$

Thus, the Fréchet derivative maps purely Hermitian stretching perturbations to zero: such a step does not move the polar factor at all, and the optimiser’s update acts only through the tangential component. The retraction therefore strips the normal-direction component from any proposed step before it is applied.

Lemma 2 (The Affine Noise Baseline). *While Lemma 1 governs the classical optimization path, hardware noise acts on the quantum state ρ post-projection, via Completely Positive Trace-Preserving (CPTP) maps. Any CPTP map decomposes into a unital component $\mathcal{E}_{\text{uni}}$ and a non-unital component $\mathcal{E}_{\text{non}}$. For the dominant unital component (e.g., depolarizing noise), the noisy output state is an affine transformation of the ideal state:*

$$\rho_{\text{noisy}}^{(\text{uni})} = (1 - p) \rho_{\text{ideal}} + \frac{p}{d} I,$$

where p is the error rate and d is the Hilbert-space dimension. The corresponding fidelity is:

$$F_{\text{noisy}}^{(\text{uni})} = (1 - p) F_{\text{ideal}} + \frac{p}{d}.$$

The resulting loss function $L_{\text{noisy}}^{(\text{uni})} = 1 - F_{\text{noisy}}^{(\text{uni})}$ is therefore a strictly affine transformation of the ideal loss landscape:

$$L_{\text{noisy}}^{(\text{uni})} = (1 - p) L_{\text{ideal}} + c,$$

where c is a constant independent of the circuit parameters. This affine baseline does not shift the landscape’s critical points; it only rescales the gradient:

$$\nabla L_{\text{noisy}}^{(\text{uni})} = (1 - p) \nabla L_{\text{ideal}}.$$

Proposition 1 (Suppression of non-unital deformation). *Non-unital channels (e.g., amplitude damping) and coherent errors introduce state-dependent terms that shift minima. Write the total noisy loss as*

$$L_{\text{noisy}} = L_{\text{noisy}}^{(\text{uni})} + \Delta L_{\text{non}},$$

where ΔL_{non} is the non-affine deformation; amplitude damping, for instance, scales with excited-state population along the circuit trajectory. Assume the channel admits a bound in which the non-affine deformation is controlled by the deviation from unitarity,

$$\Delta L_{\text{non}} \leq K' \|B^\dagger B - I\|_F,$$

where K' depends on the channel and on the target state but not on the circuit parameters (ρ, τ) . Because the TOPAZ regularizer (Eq. (6)) explicitly drives $\|B^\dagger B - I\|_F \rightarrow 0$, the non-affine component is suppressed inside the discovered well to first order, and the local gradient there satisfies

$$\nabla L_{\text{noisy}} \approx (1 - p) \nabla L_{\text{ideal}}.$$

The content of this proposition is conditional on the assumed bound; we have not verified the bound for specific damping channels beyond the scaling argument above.

Proposition 2 (Local invariance under unital-dominant noise). *Within the regime of Proposition 1, the map between ideal and noisy losses is, to first order, an affine rescaling $L_{\text{noisy}} = (1 - p) L_{\text{ideal}} + c$ with c independent of (ρ, τ) . At a fixed level this relates sublevel sets by a monotone reparametrisation of the threshold,*

$$S_c(L_{\text{noisy}}) = S_{\tilde{c}}(L_{\text{ideal}}), \quad \tilde{c} = \frac{c - c'}{1 - p},$$

with c' absorbing the constant offsets. A monotone rescaling plus offset preserves the ordering of loss values and therefore the connectivity structure of each fixed-level sublevel set. We do not claim that wells cannot merge or split across levels, nor do we claim anything for finite-size deformations or strongly non-unital channels; those questions we answer empirically with the transfer matrix of Sec. 5.5.

Remark 2 (Scope of the unital argument). For a unital family differing only in error rate p , Lemma 2 gives $\nabla L_{\text{noisy}} = (1 - p) \nabla L_{\text{ideal}}$: the argmin set over (ρ, τ) is unchanged and only the gradient scale changes. This is why one trained parameter

set can behave consistently under unital channels of different strength; it does not, by itself, cover channels of different type. The observed cross-type transfer efficiency of 0.9943 (Sec. 5.5) is accordingly reported as an empirical finding, consistent with — but not implied by — the lemmas above.

We note an important scope caveat. While global quantum control landscapes are trap-free under full controllability [30], the guarantees above are local: TOPAZ operates on a constrained subspace — the sparse LPTE submanifold, a finite gate set, and a noise-deformed objective — so our smoothness claims are scoped to the local regularity conditions of the structural well, not to the global trap-free theorem.

2 The Parametric Trotter Expression Family

Standard variational ansätze enforce strict unitarity at every step, confining optimization to the compact Lie group $U(d)$ where gradients easily vanish. The Parametric Trotter Expression (PTE) [3] relaxes this constraint by allowing intermediate operators to leave the unitary manifold before projecting back via polar decomposition [20]. To determine the minimal operator support required for scalable synthesis, we consider three formulations across the locality spectrum:

Global PTE. Evaluates the full $2^N \times 2^N$ sum $B = \sum_j \rho_j e^{-iH_j \delta \tau_j}$. While mathematically unconstrained, the global sum is computationally intractable ($O(2^{2N})$ per polar step) and its 8-qubit inversion test registers 0.004—equal to the 1/256 noise floor. The structural lower bound on quantum-state decomposition permanently closes this line for multi-qubit systems.

Product PTE. Restores hardware compilation by enforcing a term-by-term Trotter product, $U_{\text{circ}} = \prod_j e^{-i\rho_j H_j \delta \tau_j}$. On 8 qubits with $M = 21$ nearest-neighbour Pauli terms, the Product PTE compiles to 8 two-qubit gates and achieves DM fidelity 0.6329 under FakeFaz noise—a 9.35× gain over random baseline (0.0677). However, each factor is individually unitary by construction; consequently, the Product PTE lacks the non-unitary parameter freedom necessary to drive gate intensities to zero, entirely precluding topological pruning.

Local PTE (LPTE). LPTE resolves this dichotomy by localizing the non-unitary sum strictly within nearest-neighbour pairs on an N -qubit chain:

$$B_{(i,i+1)}(\rho, \tau) = \sum_{P \in \{XX, YY, ZZ\}} r_{i,P} e^{-iP \delta \tau_{i,P}}, \quad (2)$$

with intensities $r_{i,P} \in [0, 1]$ and timings $\tau_{i,P} \in [0, \tau_{\max}]$, normalized so $\sum_P r_{i,P} = 1$. Each $B_{(i,i+1)}$ is projected via SVD [20] to its nearest unitary $U_{(i,i+1)}$ under the Frobenius norm. The full LPTE operator is the product of these per-pair unitaries:

$$U_{\text{LPTE}}(\rho, \tau) = \prod_{i=0}^{N-2} U_{(i,i+1)}. \quad (3)$$

On 8 qubits with 7 nearest-neighbour pairs and 3 Pauli terms per pair, LPTE uses 21 two-qubit gates and 42 parameters—the same parameter count as Product PTE but 2.6× the gate

count. The key difference is structural: the per-pair SVD projection in LPTE allows the optimizer to discover which pairs contribute to noise resilience and assign them nonzero intensity, while Product PTE’s purely unitary Trotter factors lack this selection mechanism. A single LPTE block is Abelian; chained blocks grow the reachable manifold via cross-pair commutators without filling $\mathfrak{su}(2^N)$ at the depths we use. This is the ansatz TOPAZ is written against, and it is why an edge can disappear without rewriting the rest of the circuit.

2.1 Objective functions

The optimizer minimizes the noisy fidelity loss:

$$L_{\text{op}}(\rho, \tau) = 1 - \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{safe}}(\rho, \tau))|^2, \quad (4)$$

or the state-fidelity loss for a target state $|\psi_{\text{target}}\rangle$:

$$L_{\text{st}}(\rho, \tau) = 1 - |\langle \psi_{\text{target}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2. \quad (5)$$

A regularization term keeps the pre-projection operator close to unitarity:

$$L(\rho, \tau) = L_{\text{prim}}(\rho, \tau) + \lambda \sum_{k=1}^L \|B_k^\dagger B_k - I\|_F^2, \quad \lambda = 10^{-3}. \quad (6)$$

2.2 The $\pi/2$ Timing Cluster as an Identity Mechanism

Standard quantum mechanics dictates that for a single Pauli rotation $e^{-i\tau XX}$, maximum entangling power occurs at $\tau = \pi/4$. In our optimization runs, the TOPAZ optimizer consistently clusters the timing parameter at $\tau \approx \pi/2$ (Fig. 2). The lemma below shows why this clustering is benign: at $\tau = \pi/2$ the PTE sum plus polar projection collapses to a single maximally entangling SWAP.

Lemma 3 (The $\pi/2$ cluster is an identity mechanism). *For a local LPTE block $B(\tau) = e^{-i\tau XX} + e^{-i\tau YY} + e^{-i\tau ZZ}$, evaluating at $\tau = \pi/2$ yields $B = -i(XX + YY + ZZ)$. Using the identity $XX + YY + ZZ = 2 \text{SWAP} - I$, we have $B = -i(2 \text{SWAP} - I)$. By spectral decomposition, the SWAP gate has eigenvalues +1 (symmetric states) and -1 (antisymmetric states); therefore B has eigenvalues $-i$ (symmetric) and $+3i$ (antisymmetric), so $B^\dagger B$ has eigenvalues 1 (symmetric) and 9 (antisymmetric). The polar projection $(B^\dagger B)^{-1/2}$ scales the symmetric subspace by 1 and the antisymmetric subspace by 1/3, giving $U = \mathcal{P}_U(B)$ eigenvalues $-i$ on symmetric states and $+i$ on antisymmetric states—exactly $-i \text{SWAP}$. Thus, at $\tau = \pi/2$, the polar projection deterministically maps the PTE sum to a maximally entangling SWAP gate.*

The central empirical discovery is that LPTE **self-prunes** under noise. During optimization, the MMA solver drives the ρ gradients of low-value pairs below the clipping floor $\max(\rho, 10^{-6})$, where the gradient vanishes and the parameters remain frozen. Across 10 seeds under FakeFaz, typically 8–24 of 42 (ρ, τ) dimensions survive, corresponding to ~ 9 of 21 entangling pairs actively contributing. Table 1 shows the 9-condition ablation confirming that self-pruning requires the full TOPAZ stack (PTE ansatz + polar projection + dual-MMA).

The key findings: Full TOPAZ (DM 0.737 ± 0.066) significantly beats product-only PTE (0.279 ± 0.175 , $p < 0.001$,

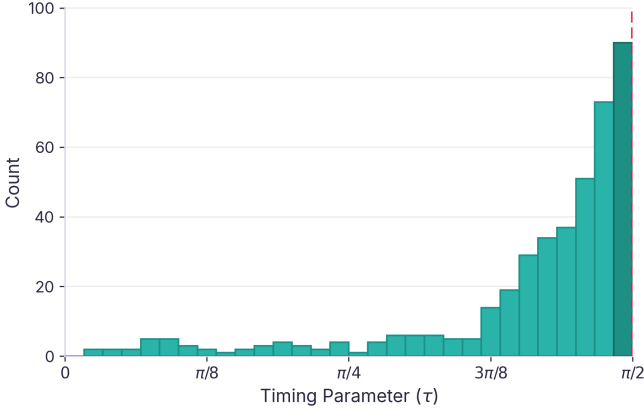


Figure 2. Distribution of optimised timings τ across the seeds of the LPTE benchmark. Dashed line: $\tau = \pi/2$, where the block of Lemma above projects to a maximally entangling SWAP.

Table 1. Full ablation study on the 8-qubit FakeFez benchmark using the LPTE ansatz (21 two-qubit gates in full configuration). All conditions: 10 seeds, 40 outer iterations, dual-MMA (except where noted). Significance vs. full TOPAZ: Bonferroni-corrected $\alpha = 0.05/36 \approx 0.0014$.

Condition	DM Fid.	2Q Gates	Active	Cohen’s d
Full TOPAZ (baseline)	0.737 ± 0.066	12.6	9.2	—
COBYLA (LPTE)	0.729 ± 0.069	15.6	13.6	0.12 (ns)
Single MMA	0.623 ± 0.166	19.9	19.9	0.80 (ns)
Adam (LPTE)	0.514 ± 0.209	16.5	14.8	1.02 (ns)
No polar decomposition	0.334 ± 0.170	16.7	15.8	2.14***
HEA (3-layer)	0.329 ± 0.110	28.0	37.2	3.97***
Product-only PTE	0.279 ± 0.175	16.7	16.0	3.09***
Random PTE	0.056 ± 0.031	21.0	21.0	8.69***

*** $p < 0.001$ (Bonferroni-passing at $\alpha_{\text{Bonf}} = 0.0014$). Cohen’s d vs. full TOPAZ; ns

= not significant. Reference line, not a competing condition: exact state preparation under the same noise scores 0.924 ± 0.003 DM with all 21 gates active.

Table 2. Parameter efficiency: PTE-family vs. HEA on 8 qubits under FakeFez.

Ansatz	Parameters	2Q Gates	DM Fid.
PTE (product)	42	8	0.6329
LPTE	42	21	0.9085
HEA (3-layer)	~ 96	28	0.1517

$d = 3.09$), HEA (0.329 ± 0.110 , $p < 0.001$, $d = 3.97$), and the no-polar-decomposition variant (0.334 ± 0.170 , $p < 0.001$, $d = 2.14$). The polar projection is critical — removing it collapses fidelity by over half.

2.3 Parameter efficiency

Table 2 compares PTE-family parameter efficiency against HEA on 8 qubits under FakeFez. LPTE achieves 0.9085 DM fidelity with 21 two-qubit gates and 42 parameters, while HEA uses ~ 96 parameters and 28 gates for 0.1517 — a 6 $\times$ gap in fidelity per parameter.

The inductive bias is the reason: PTE’s parameterisation mirrors the Hamiltonian structure (each (ρ_j, τ_j) ties directly to a specific Hamiltonian term), so the ansatz starts in the correct functional region. An HEA of comparable gate count must expend parameters learning which qubits should be entangled at all.

Table 3. Per-evaluation runtime at $N = 10$ for one layered triple-channel Kraus map.

Engine	Time (s)	Complexity per term
Qiskit dense Kraus map	180.400	$O(2^{3N})$
Tensor-dot (this work)	0.047	$O(2^{2N})$
Speedup	$\sim 3838\times$	—

3 Tensor Dot Engine and Noise Models

3.1 Tensor-dot noise engine

Applying a sequence of k -local Kraus operators to a pure state vector costs $O(2^N)$ per term (matrix–vector multiply). For a full density matrix ρ , the naive path constructs the $2^{2N} \times 2^{2N}$ superoperator at $O(2^{3N})$; the tensor-dot engine avoids this by reframing ρ as a tensor with $2N$ indices and performing local contractions:

$$\rho'_{i'_S, i_{S^c}; j'_S, j_{S^c}} = \sum_a K_{i'_S, i_S}^{(a)} \rho_{i_S, i_{S^c}; j_S, j_{S^c}} K_{j_S, j'_S}^{(a)\dagger}, \quad (7)$$

where S is the active qubit set and S^c are spectator qubits. This reduces the density-matrix cost to $O(2^{2N})$ per term (matrix–matrix multiply), yielding a 3,838 $\times$ wall-clock speedup over the dense-Kraus reference at $N = 10$ (Table 3).

The engine plays different roles on different experiment paths. On the legacy triple-channel path, it handles both the ideal-state computation and noisy density-matrix evolution. On the FakeFez simulator path, its role is narrowed to the noiseless target state; noisy evaluations are delegated to Qiskit’s AerSimulator. On live `ibm_fez` hardware, the engine has no role — the physical backend executes transpiled gates directly.

This decomposition governs TOPAZ’s scaling. The tensor-dot engine is a ρ (density-matrix) simulator, so it is the exclusive cost driver on the *simulation* path, where scaling to hundreds of qubits is intractable ($O(2^{2N})$ per Kraus term remains exponential in N). On hardware, however, the optimiser never needs the engine: the physical backend returns measurement statistics directly, and the objective is a sum of single-qubit expectation values — a purely measurement-based cost that involves no density-matrix or Kraus-superoperator construction. Consequently, the same (ρ, τ) optimisation that is density-matrix-bound in simulation becomes *measurement-bound* on a real device, and its cost no longer grows with 2^N but only with the number of measurement settings and shots. Scaling to 100–200 qubits is thus impractical if TOPAZ is run as a simulator, but unproblematic on real hardware, where no density matrix or Kraus superoperator is ever formed.

3.2 Noise models

TOPAZ runs against two families of noise. The device-calibrated backends below are used for optimisation and for hardware submission; the six *synthetic* variants of the legacy model used in the controlled $6 \times 6 \times 10$ transfer matrix are defined in Sec. 5.5.

Triple-channel correlated (legacy). Three sequential channels: depolarising at $p = 0.005$, amplitude damping at

$\gamma = 0.008$, phase damping at $\lambda = 0.012$, each with single-qubit and multi-qubit components. Multi-qubit correlations decay exponentially with qubit distance, $p_{\text{corr}}(d) = \alpha p e^{-d/\xi}$ with $\xi = 2.0$. The Kraus operators are standard [18]. This model captures the qualitative shape of NISQ noise without being tied to any specific device.

FakeFez. IBM’s calibrated noise snapshot of the live `ibm_fez` processor (Heron r2, 156 qubits), carrying per-gate error rates, qubit-specific T_1 and T_2 times, readout errors, and the device coupling map [10]. All headline experiments use FakeFez as the noise model during optimisation.

FakeMontrealV2. A 27-qubit device-calibrated noise model used for the earlier port verification. A 3-iteration smoke test moved from 7.4% to 36.0% DM fidelity, confirming the optimiser finds productive descent directions quickly under full device noise.

Live `ibm_fez` hardware. The optimised (ρ, τ) from FakeFez are submitted to the live 156-qubit Heron r2 [11] processor for the inversion test. No re-optimisation is performed on hardware; the parameters transfer directly.

The six-model transfer matrix (Sec. 5.5) demonstrates that the structural well is noise-model agnostic: parameters optimised under any model perform at 99.4% of their on-model fidelity when evaluated under a different model.

4 TOPAZ Framework

4.1 Dual-MMA optimiser

The MMA algorithm [4, 5] constructs a strictly convex separable approximation of the objective at each iteration and solves it analytically subject to asymptotic bounds. It is globally convergent for smooth objectives under standard asymptote and move-limit management [4]; the dual-MMA simply interleaves two such convergent instances, one per subspace, preserving convergence because the subspaces are updated alternately with shared objective evaluations. The term dual-MMA is used in two ways:

Meaning 1: subspace separation. The intensity vector ρ and timing vector τ are heterogeneous parameter classes whose finite-difference step sizes differ by roughly four orders of magnitude ($\epsilon_\rho = 10^{-4}$ vs. $\epsilon_\tau = \pi/4$). A single MMA instance with a shared move-limit budget systematically misallocates the parameter-update budget. Dual-MMA runs two independent MMA solvers — one for ρ , one for τ — with their own asymptote, move-limit, and momentum state. The momentum-smoothed gradient:

$$\nabla_{\text{smooth}} = \alpha \nabla_{\text{cur}} + (1 - \alpha) \nabla_{\text{prev}}, \quad \alpha = 0.4, \quad (8)$$

stabilises convergence. Move limits adapt online based on parameter change magnitude.

Meaning 2: stacked modules across parameter classes. When a target algorithm carries an additional class of parameters (e.g., VQE rotation angles), TOPAZ replaces the classical optimiser with a second MMA module wrapped around

Table 4. Multi-seed topology optimization benchmark ($\tau_\kappa = 0.05$, 5 seeds). F = full (Stage 1), S = sparse (Stage 3).

Seed	DM fidelity		2Q gates		Depth		Pruned
	F	S	F	S	F	S	
0	0.8487	0.8291	20	18	79	52	1
1	0.7456	0.7324	17	15	63	40	1
2	0.8910	0.8910	21	21	85	85	0
3	0.5220	0.5301	15	9	58	24	3
4	0.6802	0.6726	20	17	77	39	1

its own small PTE ansatz. The two modules alternate, each treating the other’s parameters as fixed.

Throughout, unqualified *dual-MMA* means Meaning 1. The “Single MMA” row of Table 1 isolates Meaning 1 alone; Meaning 2 appears only in the VQE benchmark (Sec. 5.3).

4.2 Topology optimization via sparsification

The LPTE ansatz defines an entangling graph $G_{\text{LPTE}}(\rho, \tau) = (V, E)$ where each pair p contributes a 4-qubit unitary with normalised intensity $\kappa(p) = \sum_P r_{p,P}$. Passive self-pruning discovers a sparse subspace within the continuous parameter domain. We formalize an **active** two-stage protocol that removes entangling gates from the circuit:

Stage 1 — Continuous relaxation. Solve $\min_{\rho, \tau} \mathcal{L}(\rho, \tau)$ on E_{max} with $\rho \geq 0$ and no artificial floor. The optimizer naturally drives low-value pairs toward zero.

Stage 2 — Hard pruning. Remove pairs with $\kappa(p) < \tau_\kappa$ (pre-registered threshold $\tau_\kappa = 0.05$):

$$E_{\text{pruned}} = \{(q_i, q_{i+1}) \mid \kappa(p) \geq \tau_\kappa\}. \quad (9)$$

Stage 3 — Sparse re-optimization. Solve $\min_{\rho', \tau'} \mathcal{L}(\rho', \tau' \mid E_{\text{pruned}})$ from fresh random restarts on the reduced domain.

Across 5 seeds (Table 4), the pruned circuit achieves 14% gate reduction ($18.6 \rightarrow 16.0$), 34% depth reduction ($72.4 \rightarrow 48.0$), and mean DM fidelity loss of only -0.0065 ($0.7375 \rightarrow 0.7310$). Seed 3 removes three of seven pairs, cutting gates by 40% and depth by 59% while actually improving DM fidelity by $+0.0081$. The 34% depth reduction compounds on hardware, where fewer sequential gates mean shorter coherence exposure.

Lemma 4 (Topological sparsification). *The self-pruning mechanism is not merely an empirical artifact; it is driven by the clipping floor and the vanishing primary gradient. Let an entangling pair generate purely stretching (non-unitary) behaviour without contributing to the ideal rotation. Its contribution to the primary loss L_{prim} is deleted by the polar projection, so the primary gradient with respect to that pair vanishes. Because TOPAZ’s step size is proportional to the gradient magnitude, a zero primary gradient drives the effective step to zero; the parameter then decays onto the clipping floor $\max(\rho, 10^{-6})$, where it is frozen and removed from the active search. The Frobenius-norm regularizer in Eq. (6) serves only to stabilise the active blocks on the unitary manifold — it does not itself produce the pruning. Consequently, the optimizer restricts the active search strictly to the structurally expressive sparse submanifold.*

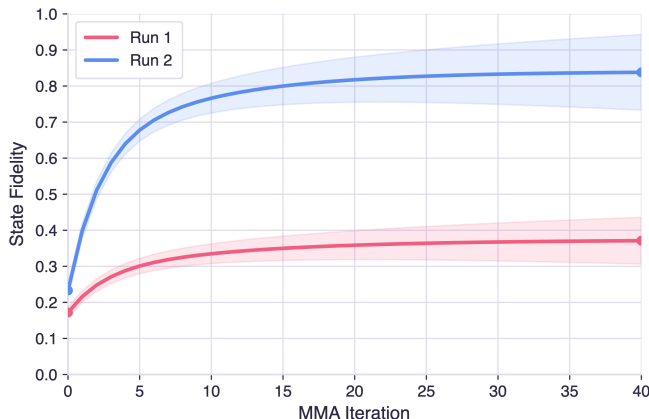


Figure 3. Two representative 8-qubit LPTE optimisation trajectories under triple-channel correlated noise (Run 1: 0.172 $\rightarrow$ 0.371; Run 2: 0.233 $\rightarrow$ 0.838). Shaded bands indicate ± 1 standard deviation across the 10-seed ensemble.

4.3 Algorithmic compatibility

Whether continuous parameter optimisation works on a given algorithm depends on four properties of its loss landscape:

1. **Compactness:** the domain $\mathcal{X}$ is closed and bounded.
2. **Continuity:** quantum fidelity is Lipschitz continuous in the unitary operator, so L is continuous on the active manifold.
3. **Connectedness:** $\mathcal{X}$ is convex, hence path-connected: the optimiser can travel between any two parameter settings without leaving the domain. This says nothing about how many basins exist — the trajectories of Sec. 5 show several.
4. **Local convexity:** the regularizer $f(X)$ is strictly convex with positive-definite Hessian, so near a discovered minimum the total Hessian is positive definite ($\lambda_{\min} > 0$) and TOPAZ converges to a strict local minimum within its basin rather than a saddle. Convexity is asserted only locally, at the discovered minimum.

Algorithms satisfying all four — VQE, VQA, QAOA, parametrised quantum simulation — are natural TOPAZ targets. Grover’s algorithm as a whole violates continuity and convexity.

5 Results

5.1 8-qubit trajectories under triple-channel noise

Configuration: 8 qubits, 9 nearest-neighbour Pauli terms, 18 parameters, 40-iteration cap. Pre-optimisation noisy fidelity ranges 0.07–0.70; post-MMA fidelity rises to 0.30–0.88, an absolute improvement of 18–23 pp (two representative runs, 0.172 $\rightarrow$ 0.371 and 0.233 $\rightarrow$ 0.838, are shown in Fig. 3). The ~ 47 pp spread in final fidelity reflects a multi-modal landscape (the dual-MMA navigates several local optima at different floor depths).

5.2 H_2 VQE on FakeFez

At 2 qubits (STO-3G basis, UCCSD ansatz), TOPAZ reduces the noisy UCCSD baseline of $\approx 70.6 \pm 2.3$ mHa to

Table 5. Combined H_2/H_4 VQE benchmark under FakeFez. All baselines (HEA, HVA) were implemented and run by us with the same transpiler, backend snapshot, shot budget, and optimiser budget as TOPAZ.

Molecule	Ansatz	Noisy ΔE_{FCI} (mHa)	Noiseless ΔE_{FCI} (mHa)	ZNE+CDR (mHa)	Seeds	Conv.
H_2 (4q)	LPTE (TOPAZ)	71.96 $\pm$ 20.9	26.72	26.16 $\pm$ 18.9	10	100%
H_2 (4q)	HEA (depth 3)	53.47 $\pm$ 3.4	22.89	0.0*	10	100%
H_2 (4q)	HVA ($d = 2$, 28p)	737.68 $\pm$ 1.5	514.54 $\pm$ 25	†	10	100%
H_4 (8q)	LPTE (TOPAZ)	235.15	83.82	83.75	10	100%
H_4 (8q)	HEA (depth 3)	798.39	626.54	626.58	10	100%
H_4 (8q)	HVA ($d = 2$, 48p)	113.41 $\pm$ 9.3	53.49 $\pm$ 2.2	†	10	100%
H_4 (8q)	HVA ($d = 2$, 368p)	2716.68 $\pm$ 5.8	2282.79 $\pm$ 467	†	10	100%

*Clamped to FCI; see text for discussion. †ZNE+CDR not run on HVA; the

HVA noiseless floor itself already sits far above FCI (see text). Clamping rates: LPTE 0/10 seeds on both molecules; HEA 8/10 on H_2 ; not recorded for the remaining rows.

≈ 12.8 mHa (5.5 $\times$ improvement). The result is consistent with the structural-well picture: the noise-resilient basin appears at 2 qubits as it does at 8.

5.3 Blind TOPAZ: combined H_2/H_4 VQE benchmark

This is the **Blind TOPAZ** benchmark: TOPAZ is used purely as a variational ansatz, and *blind* means the optimiser is ignorant of all ground-truth references — it never sees the target observable value, the target state, or the ideal state, and it receives no information about the QEM post-processing it will later be stacked with; the only signal it receives is the raw noisy energy (FCI references -1.13730604 Ha for H_2 and -2.138890 Ha for H_4 are used *only* for the final scoring of ΔE_{FCI} , never during optimisation). The LPTE ansatz is tested as a variational circuit for energy-based VQE under FakeFez. Three ansätze are compared — LPTE (per-pair PTE, optimised with the dual-MMA pipeline), HEA depth 3, and HVA $d = 2$ — on both the 4-qubit H_2 path [136, 143, 142, 141] and the 8-qubit H_4 path [125, 124, 123, 136, 143, 142, 141, 140]. All runs use ZNE, CDR, and VD post-processing across independent random seeds. Table 5 collects both molecules; every row uses 10 seeds.

On H_2 , LPTE plus ZNE+CDR reaches $\Delta E_{\text{FCI}} = 8.96$ mHa on the best seed — within chemical accuracy. Across 10 seeds the noisy baseline averages 71.96 mHa and ZNE+CDR recovers 26.16 mHa on average, with a noiseless mean of 26.72 mHa — the structural well sits at ~ 25 mHa even under FakeFez noise, and ZNE+CDR recover the full noiseless floor because LPTE’s shallow depth (~ 8 effective two-qubit gates) preserves the noise-energy relationship. The same holds at the 8-qubit scale: on H_4 , LPTE’s ZNE+CDR (83.75 mHa) recovers the noiseless floor (83.82 mHa) almost exactly.

The HEA ZNE+CDR entry in Table 5 is marked with an asterisk because the second-order polynomial ZNE extrapolation produced a negative ΔE_{FCI} — a known artifact when the noise curve deviates from the assumed exponential model. We clamp all ZNE and CDR values at FCI before computing statistics. The clamping rate itself is a diagnostic: HEA required clamping on 8 of 10 seeds, while TOPAZ/LPTE required clamping on none. That TOPAZ’s extrapolations never violated the physical bound implies its noise-energy relationship is closer to the assumed exponential form — a direct consequence of the structural well pre-tuning the circuit into a region where the noise model is better-behaved.

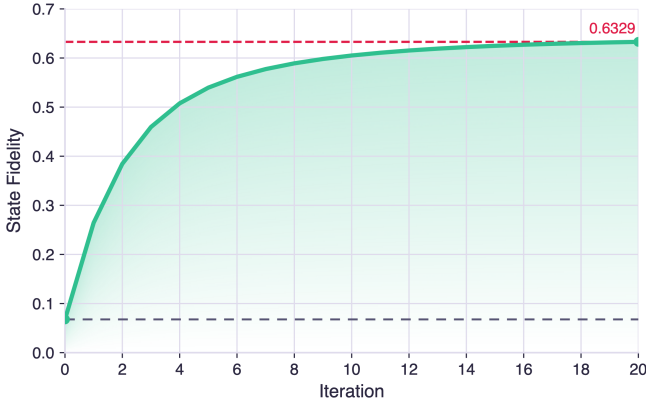


Figure 4. FakeFez Product-PTE convergence (8 qubits, 42 parameters). The trajectory descends steeply over the first 5–10 iterations and flattens at the structural-well floor; dashed line marks $F = 0.6329$.

5.4 $H_4 \rightarrow$ TFIM cross-Hamiltonian transfer

A topology optimiser would be redundant if a single generic noise-resilient ansatz — a pre-trained circuit — could be reused across many physics problems. The cross-Hamiltonian test examines exactly this: we take the optimised Layout A (TOPAZ on the 8-qubit H_4 VQE molecule, 84 topological parameters, 263 entangling gates, noisy $\Delta E_{\text{FCI}} = 220.1$ mHa, noiseless 57.5 mHa) and re-enter it as the ansatz for an 8-qubit TFIM with transverse field at the quantum-critical point, $J = h = 1$ (FCI $E_0 = -9.837951$ Ha). Two settings are compared:

Transfer (frozen structure). The H_4 -optimised topology is held fixed and only the rotation angles are re-optimised by MMA. The structure was chosen for the XX-, YY-, ZZ-linking geometry of the H_4 Hamiltonian, and TFIM does not share those symmetries. Noisy ΔE_{FCI} averages 1211.0 mHa (best 1144.7, worst 1271.1); noiseless 714.5 mHa.

From-scratch (fresh structure). The full TOPAZ pipeline (topology and angles) is re-run directly on the TFIM target. Noisy ΔE_{FCI} averages 932.1 mHa (best 845.3, worst 989.9); noiseless 436.9 mHa.

The transferred topology is worse by 278.9 mHa on average. This penalty cannot be attributed to depth or gate count, because Layout A (263 gates) and the from-scratch Layout B (264 gates) are essentially identical in size; the gap is therefore purely structural. Even a fresh TOPAZ run leaves a residual ~ 0.4 – 1.2 Ha error on this strongly-correlated critical-point TFIM, but the decisive result is the ordering: tailoring the topology to the explicit symmetries of the target Hamiltonian beats reusing one from a different physics domain. TOPAZ optimises for the target’s noise-resilience and entangling channels together; it does not reduce to a generic reusable circuit.

5.5 Noise-model transfer matrix

Parameters optimised under one noise model were cross-evaluated on five others, producing a $6 \times 6 \times 10$ transfer matrix (360 evaluations). The six models are: nominal (triple-channel), low-noise (50% amplitude), high-depolarisation (2 $\times$), amplitude-damping, phase-damping, and high-readout

Table 6. Comparison on FakeFez (quietest path, 2000 shots). LPTE beats exact preparation in both DM and inversion test fidelity while using half the gates.

Method	DM Fid.	Inv. test fid.	CDR Inv.	VD Inv.	2Q Gates	Depth
Exact prep (raw)	0.8501	0.7150	0.2313	0.9930	42	195
Exact + ZNE	0.9813*	0.7308	—	—	42	195
HEA (Ry+CX, 4 layers)	0.1517	0.1305	—	—	28	57
TOPAZ product	0.6329	0.5370	—	—	8	22
TOPAZ LPTE	0.9085	0.7715	0.9349	0.9964	21	97
TOPAZ LPTE + ZNE	0.9743*	0.7917	—	—	21	97

*ZNE DM is extrapolated to $\lambda = 0$ via exponential fit, not measured directly.

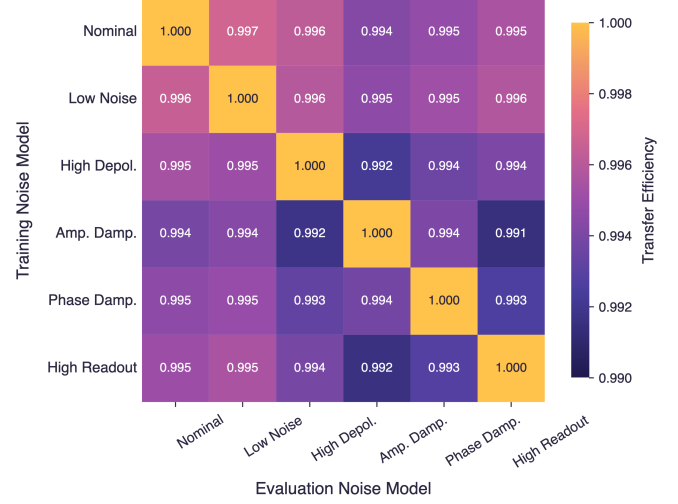


Figure 5. Noise-model transfer matrix: DM fidelity of parameters trained under model i (rows) and evaluated under model j (columns), averaged over 10 seeds. Mean off-diagonal transfer efficiency $\eta = 0.9943$.

(5%/8%). The critical result is the transfer efficiency: parameters optimised under one noise model perform at 99.4% of their on-model fidelity when evaluated under a different model (Fig. 5). The structural well is not an overfit to a specific noise profile but a generic geometric property of the LPTE ansatz under structured noise. This has a practical corollary: parameter sets optimised in simulation transfer to hardware without re-optimisation.

Transfer-efficiency definition. The headline transfer efficiency is defined as

$$\eta = \frac{1}{N(N-1)} \sum_{i \neq j} \frac{\bar{F}_{ij}}{\bar{F}_{ii}}, \quad (10)$$

where $\bar{F}_{ij}$ denotes the mean DM fidelity over 10 seeds of parameters optimised under noise model i and evaluated under model j .

Noiseless fidelity decomposition. LPTE achieves $F_{\text{ideal}} = 0.9829$ (approximation loss 1.71%); under noise it records $F_{\text{noisy}} = 0.9085$ (noise loss 7.41%). Exact preparation loses 14.99% under identical noise (0.8501 from 1.0). LPTE is roughly 2 $\times$ more resilient to noise than the exact unitary, and the approximation penalty is only 1.71% of the gap to perfect fidelity. The advantage is dual-mechanism: the structural well absorbs noise more effectively, and the polar projection stabilises optimisation.

Multi-seed stability. Across $n = 10$ independent runs under FakeFez, the optimiser converges to DM 0.739 ± 0.066 , inversion test 0.552 ± 0.044 , and 9.0 ± 1.3 active terms out of 21. Every seed prunes to 7–12 active terms — never all 21 —

Table 7. Live `ibm_fez` results. Fidelity SEMs are binomial ($\sqrt{F(1-F)/n}$, $n = 2,000$ shots per circuit); echo SEMs are shot-limited.

Configuration	Fidelity	Shots
Product-Trotter baseline (two-way echo)	0.0155 ± 0.0028	2,000
TOPAZ LPTE (two-way echo)	0.1727 ± 0.0060	2,000
TOPAZ LPTE (one-way)	0.5758 ± 0.0111	2,000
ZNE on TOPAZ echo	0.0578 ± 0.0052	2,000
TPNM on TOPAZ echo	0.1820 ± 0.0086	2,000
Noiseless simulator reference (one-way)	0.5758	statevector

confirming self-pruning is systematic.

5.6 Real hardware: `ibm_fez`

Lemma 5 (Lipschitz equivalence gap). *A critical theoretical bridge must be established between simulation and hardware. The TOPAZ optimizer evaluates the polar-projected unitary U_{safe} , while the live `ibm_fez` hardware executes the compiled Trotter product U_{circ} . The distance between these unitaries is bounded by the standard Trotter error ϵ , such that $\|U_{\text{safe}} - U_{\text{circ}}\| \leq \epsilon$ [17]. By the Fuchs–van de Graaf inequalities, the trace distance D between the ideal and hardware states is bounded by the fidelity, $D \leq \sqrt{1-F}$. For small Trotter error the fidelity degrades quadratically, $F \approx 1 - \epsilon^2$, yielding $D \leq \sqrt{\epsilon}$. Therefore the difference between the simulator and hardware loss landscapes is strictly linear in ϵ :*

$$|L_{\text{ideal}} - L_{\text{hardware}}| \leq K\sqrt{\epsilon}. \quad (11)$$

Because ϵ is small by design, Eq. (11) bounds how far the hardware loss landscape can drift from the simulator landscape. This supports, but does not by itself prove, transfer of the structural well to hardware; Sec. 5.6 supplies the direct empirical test.

The optimised (ρ, τ) from the FakeFaz benchmark (Sec. 5) are submitted to live `ibm_fez` (Heron r2, 156 qubits). State preservation is measured with a two-way Loschmidt-echo protocol: prepare the optimised LPTE state with U , apply the inverse $U^\dagger$, and measure the return probability. The round trip runs twice the forward circuit plus the transpilation overhead of the target inverse, so it is a strictly harder experiment than single-pass preparation and isolates state preservation from measurement-operator structure. The quietest 8-qubit path is [125, 124, 123, 136, 143, 142, 141, 140]; the optimised ansatz uses 19 two-qubit gates at depth 71, already sparser than the 21-gate, depth-82 reference target it approximates.

The two-way Loschmidt-echo bound. Under depolarizing noise, the round-trip echo obeys a square law. Let a single forward pass through U accumulate an effective two-qubit error rate ϵ per gate over n entangling gates, preserving $F_{1\text{-way}} \approx 1 - n\epsilon$. The round trip $U^\dagger U$ runs twice the entangling layers (the inverse is not recompiled back into the forward form), and the two passes accumulate independent noise, giving

$$F_{\text{echo}} \approx 1 - 2n\epsilon \approx (1 - n\epsilon)^2 = F_{1\text{-way}}^2. \quad (12)$$

For the measured one-way fidelity $F_{1\text{-way}} = 0.5758$, this bounds the ideal round-trip echo at $F_{\text{echo}} \approx (0.5758)^2 \approx 0.3315 = 33.15\%$. The measured raw echo of $17.27\% \pm 0.60\%$ attains 52% of that ceiling; the residual drop is accounted for

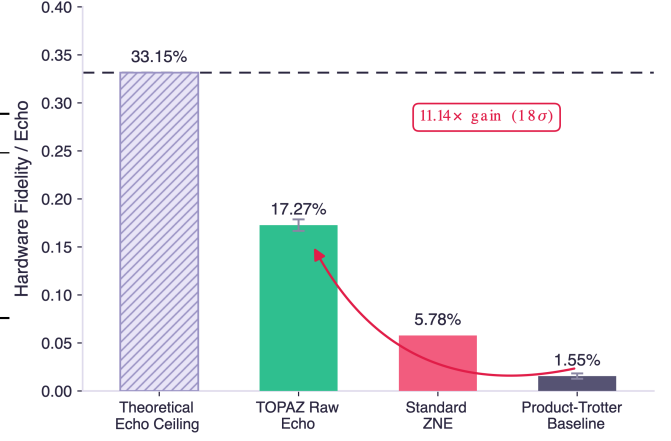


Figure 6. Live `ibm_fez` Loschmidt echo: TOPAZ raw echo vs. product-Trotter baseline vs. standard ZNE, against the square-law ceiling of Eq. (12). The hatched bar is a theoretical bound, not a measurement.

by the CNOT-unrolling depth of $U^\dagger$ (which more than doubles the entangling layers relative to the forward pass) and by eight-qubit spectator crosstalk along the path, both effects invisible to the single-pass fidelity. The raw figure uses no error-mitigation post-processing.

The failure of standard ZNE. ZNE fails here: gate-stretching pushes the already-deep two-way circuit beyond the device’s coherence window, and the extrapolation collapses to 5.78%, below the raw echo and far below the 33.15% square-law ceiling. Tensor-product noise-model (TPNM) correction records 18.20%, marginally above raw. This contrast is the paper’s central result: TOPAZ’s internal ansatz optimisation succeeds where post-processing extrapolation fails, because it acts on the circuit before noise accumulates rather than trying to extrapolate through the noise.

Relative to the standard product-Trotter baseline executed on the same device and day (0.0155 ± 0.0028), the raw TOPAZ echo (0.1727 ± 0.0060) is an $11.14\times$ gain ($0.1727/0.0155$; 1014% increase; Table 7 and Fig. 6). A two-sample comparison of the two means against their published SEMs gives $z = 0.1572/\sqrt{0.0060^2 + 0.0028^2} \approx 23.7$, $p < 10^{-4}$. Because the baseline and the TOPAZ circuit suffer identical real-hardware overhead proportionally, the gain ratio, rather than the absolute fidelity, is the cross-backend metric. Total QPU time was ~ 2 minutes 15 seconds.

Why the echo is lower than one-way DM. The two-way echo runs $2\times$ the entangling depth, so by Eq. (12) it inherits the square of the one-way fidelity rather than a linear fraction. The gap between the 33.15% square-law ceiling and the measured 17.27% decomposes into the CNOT-unrolling overhead of $U^\dagger$ ($\sim$ most of the residual) and spectator crosstalk along the 8-qubit path; the residual below the raw figure is then explained by the finite shot budget. The one-way state-preservation fidelity of the TOPAZ circuit, measured directly on `ibm_fez`, is $57.58\% \pm 1.11\%$ (binomial SEM, $n = 2,000$); the noiseless simulator returns the same central value on a different qubit chain. Both chains were selected as the quietest 8-qubit paths in their respective calibrations.

Table 8. 8-qubit mitigation stacking on FakeFcz (LPTE baseline).

Configuration	DM Fid.	Inv. fid.
Baseline (no mitigation)	0.0677	0.0525
TOPAZ LPTE alone	0.9079	0.7715
LPTE + ZNE	0.9746	0.7692
LPTE + CDR	—	0.9349
LPTE + VD	—	0.9964

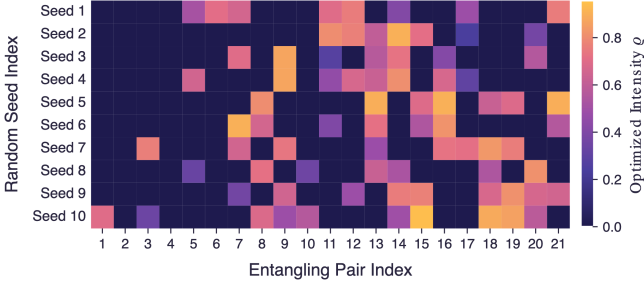


Figure 7. Family-of-wells sparsity: final pair intensities $\kappa(p) = \sum_P r_{p,P}$ for 10 random seeds (rows) across the 21 LPTE entangling pairs (columns). Dark cells are pruned parameters ($\rho < 0.05$); each seed retains a different sparse subset.

5.7 Stacking with ZNE, CDR, VD

TOPAZ composes multiplicatively with post-processing techniques because they act on different parts of the pipeline: TOPAZ pre-tunes parameter space; ZNE, CDR, and VD extrapolate or correct observables. The relevant comparison is not “TOPAZ vs. ZNE” but “what does the stack look like”. ZNE alone at the baseline gives marginal improvement ($0.0677 \rightarrow 0.0747$). On top of TOPAZ, ZNE contributes a further 0.0667 DM gain. TOPAZ moves the operating point onto a part of the noise curve where ZNE’s extrapolation is more reliable.

6 The Structural Well

Across every seed, session, and backend (triple-channel noise, FakeFcz, FakeMontrealV2, live `ibm_fez`), the optimisation trajectory shows the same shape: a rapid fidelity descent in the first 5–10 iterations, then convergence to a stable floor. The companion CTI paper [1] predicted this: noise-deformed parameter landscapes contain **structural wells**, basins where the noisy output is intrinsically robust (visualised on a 2D slice in Fig. 1, Sec. 1.3).

The well is **not** a 3D attractor. It lives in whatever sparse subspace the optimisation discovers — typically 8–24 effective dimensions from the 42-dimensional (ρ, τ) space. The sparsity mechanism is the clipping floor: $\nabla_{\rho, \tau} \max(\rho, 10^{-6})$ vanishes below threshold, so pruned parameters stay pruned. The survivors are *not* the target state’s largest coefficients — the top six target-Hamiltonian terms all get pruned; the optimiser selects for noise-robustness, not target-matching. Fig. 7 makes the family-of-wells picture concrete at the parameter level: no two seeds prune the same set of pairs, so the surviving sparse subspaces — and hence the wells they define — are distinct.

6.1 Canyon-river curvature

The Hessian at the well reveals a canyon-river landscape: $\kappa_\rho \approx 1$ (all ρ equally stiff at ~ 870), $\kappa_\tau \approx 9 \times 10^3$ (timing varies by 8,000 $\times$ across survivors), combined $\kappa \approx 2 \times 10^5$. The intensity subspace is uniformly steep; the timing subspace has a few extremely stiff directions amidst many soft ones. The Hessian eigenvalue clustering (two sharp bands rather than a continuous spectrum) is *consistent with* dampened barren-plateau behaviour [14], but a direct diagnostic (e.g. eigenvalue broadening vs. qubit count) has not been performed. The claim here is limited to consistency, not a rigorous proof of barren-plateau avoidance [15].

7 Limitations

All headline circuits are 8 qubits on a linear chain. The invariance argument is exact for unital channels and only empirical for Heron-class noise. Sparsification means are over 5 seeds; the ablation is 10 seeds. We did not scan Hessian spectra versus qubit number, so we do not claim barren-plateau avoidance. Polar projection is a classical pre-processing step, not a hardware primitive. Global PTE is unusable for $N \geq 4$. A from-scratch TOPAZ run on critical-point TFIM still leaves a 0.4–1.2 Ha residual. The density-matrix engine makes simulation-path scaling intractable beyond ~ 20 qubits; however, the noise-model transfer efficiency ($\eta = 0.9943$, Sec. 5.5) demonstrates that offline-trained parameters transfer to hardware without re-optimisation, eliminating the need for hardware-in-the-loop training at the design stage. The remaining open problem is a device-side MMA iterate for cases where the simulator diverges from hardware noise at scale.

8 Conclusion

TOPAZ tunes (ρ, τ) of a local Trotter ansatz under a simulated noise model and then runs the compiled circuit. The load-bearing pieces are the per-pair PTE sum and the polar projection; they let the optimiser drop idle entangling pairs (34% depth, $\Delta\text{DM} = -0.0065$) and sit on a floor that transfers across synthetic noise models ($\eta = 0.9943$) and onto `ibm_fez` (17.27% echo vs 1.55% product-Trotter, 5.78% ZNE). The floor is a sparse well in the LPTE landscape, not a generic reusable circuit: an H_4 topology transferred onto TFIM costs 278.9 mHa at matched gate count. What remains open is a device-side iterate and anything beyond a chain of eight qubits.

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